AMIT VERMA

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Summary

Final-year Computer Science student specializing in Machine Learning, Deep Learning, Computer Vision, and Natural Language Processing. Experience includes a DRDO Solid State Physics Laboratory internship focused on real-time thermal image processing systems. Author of a research publication accepted at ICSSAI 2025 and participant in the Amazon Machine Learning Challenge with hands-on experience developing ML pipelines and interpretable models.

Technical Skills

Programming
Python, C++, JavaScript
Machine Learning
Supervised Learning, Feature Engineering, Model Evaluation, Hyperparameter Tuning
Deep Learning
PyTorch, CNN Architectures, Transformers, GradCAM
Computer Vision
OpenCV, MediaPipe, Image Processing, Object Tracking
Tools
Git, PyQt

Experience

Machine Learning Intern Feb 2025 – May 2025
DRDO — Solid State Physics Laboratory (SSPL), Delhi
DRDO Project Executive Lab, Techno India University

- Evaluated and applied **50+ image filtering and denoising techniques** to enhance infrared image clarity and reduce noise artifacts.
- Integrated deep learning based denoising methods to improve visibility of low-bit thermal imagery.
- Architected a **multithreaded video processing pipeline** separating frame capture, filtering, and rendering to improve responsiveness and reduce CPU utilization.
- Applied **Non-Uniformity Correction (NUC)** filtering and MATLAB-based processing for **5D thermal sensor data** to improve calibration accuracy.
- Improved system stability by resolving runtime crashes and optimizing memory usage within the processing pipeline.

Education

B.Tech Computer Science
Techno India University Aug 2022 – Apr 2026
CGPA: 8.1 / 10

Projects

Brain Tumor Classification with GradCAM
GitHub

- Trained CNN-based classifier on a dataset of **3,200+ MRI brain scans across four tumor classes**.
- Applied **GradCAM and GradCAM++ explainability techniques** to visualize tumor regions influencing model predictions.
- Assessed model behavior using classification accuracy and heatmap-based interpretability analysis.

Sentiment Analysis using Transformer Models
GitHub

- Developed an **NLP sentiment classification pipeline** using Transformer architectures with HuggingFace libraries.
- Fine-tuned a pretrained model achieving **95% accuracy** and evaluated performance using precision, recall, and F1-score metrics.

Face Cursor Movement using Facial Landmarks
GitHub

- Created a **real-time cursor control system** using MediaPipe facial landmark tracking.
- Enabled nose-based cursor navigation with **wink-triggered click detection**.
- Improved responsiveness through a **multithreaded webcam processing pipeline**.

AIMLverse – Machine Learning Education Platform
aimlverse.in

- Developed an interactive ML education platform demonstrating deep learning concepts through visual simulations.
- Designed modules illustrating **backpropagation, normalization techniques, and neural network decision boundaries**.

Key Achievements

Research Publication — ICSSAI 2025
Conducted a comparative study of classical and modern image denoising algorithms across noisy datasets using quantitative image quality metrics such as PSNR and SSIM. The work was accepted for presentation at the *International Conference on Smart Systems and Artificial Intelligence (ICSSAI 2025).*

Amazon Machine Learning Challenge
Developed supervised learning pipelines involving feature preprocessing, training experimentation, and performance benchmarking.
Final Rank: 1570 ▪ **Score: 51.20**
GitHub Repository